WORKED SOLUTION

STEP III 1987, Question 6

Pure • Coupled Differential Equations

STEP III 1987, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q6

The key idea

A pair of coupled first-order equations is best turned into a single second-order equation for one unknown. The first equation lets us write y in terms of x and $\mathrm{d}x/\mathrm{d}t$, and substituting into the second leaves a forced oscillator equation for x. The whole character of the answer then depends on how the natural frequency of that oscillator compares with the frequency of the forcing $2\cos t$.

Solution

Write dots for derivatives with respect to t.

Eliminating y

From the first equation,

$$y = \tfrac{1}{5}\left(\dot{x} + 2x\right), \qquad \text{so} \qquad \dot{y} = \tfrac{1}{5}\left(\ddot{x} + 2\dot{x}\right).$$

Substituting both into the second equation and multiplying by 5,

$$\begin{aligned} \ddot{x} + 2\dot{x} + 5ax - 2\left(\dot{x} + 2x\right) &= 10\cos t, \\ \ddot{x} + (5a - 4)\,x &= 10\cos t. \end{aligned}$$

The $\dot{x}$ terms cancel, so there is no damping: this is a pure forced oscillator with natural frequency $\sqrt{5a-4}$ (when $5a > 4$).

Translating the initial conditions

We are told $x(0) = 0$ and $\dot{y}(0) = 0$, but the equation for x needs $x(0)$ and $\dot{x}(0)$. Put $t = 0$ into the second original equation:

$$\dot{y}(0) + a\,x(0) - 2y(0) = 2 \quad \Longrightarrow \quad 0 + 0 - 2y(0) = 2 \quad \Longrightarrow \quad y(0) = -1.$$

Then the first equation at $t = 0$ gives $\dot{x}(0) = 5y(0) - 2x(0) = -5$. Note that neither step used the value of a, so

$$x(0) = 0, \qquad \dot{x}(0) = -5 \qquad \text{for every } a.$$

Ask yourself The condition is on $\mathrm{d}y/\mathrm{d}t$, not on y. How do we use it?

The differential equations themselves link the values of x, y and their derivatives at any single instant. At $t = 0$ the second equation turns the information about $\dot{y}(0)$ into information about $y(0)$, and the first equation then turns that into $\dot{x}(0)$, which is exactly what a second-order equation for x needs.

The case $a = 1$

Now the equation is

$$\ddot{x} + x = 10\cos t.$$

The complementary function is $A\cos t + B\sin t$. The forcing $\cos t$ is itself a solution of the homogeneous equation (this is resonance), so a trial $C\cos t$ fails and we try $x_p = t\,(C\cos t + D\sin t)$. Differentiating twice,

$$\ddot{x}_p + x_p = -2C\sin t + 2D\cos t,$$

so $C = 0$ and $D = 5$, giving $x_p = 5t\sin t$. Hence

$$x = A\cos t + B\sin t + 5t\sin t.$$

The condition $x(0) = 0$ gives $A = 0$. Then $\dot{x} = B\cos t + 5\sin t + 5t\cos t$, so $\dot{x}(0) = B = -5$. Therefore

$$\boxed{x = 5(t-1)\sin t}$$

For y, use $y = \frac{1}{5}(\dot{x} + 2x)$ with $\dot{x} = 5\sin t + 5(t-1)\cos t$:

$$y = \sin t + (t-1)\cos t + 2(t-1)\sin t, \qquad \text{that is} \qquad \boxed{y = (t-1)\cos t + (2t-1)\sin t}$$

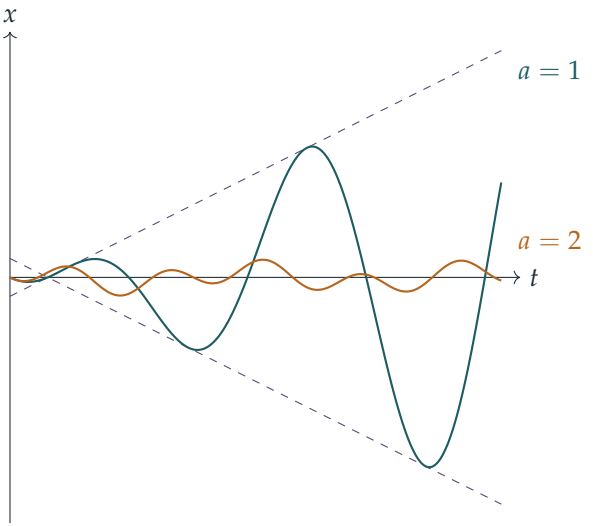
As a quick check, $y(0) = -1$, and $\dot{y} = \cos t - (t-1)\sin t + 2\sin t + (2t-1)\cos t$ gives $\dot{y}(0) = 1 - 1 = 0$, as required.

The case $a > 1$

Now $5a - 4 > 1$, so write $\omega = \sqrt{5a-4}$ with $\omega > 1$. The equation $\ddot{x} + \omega^2 x = 10\cos t$ is no longer at resonance, because the forcing frequency 1 differs from the natural frequency ω. So:

- The complementary function is $A\cos\omega t + B\sin\omega t$, oscillating with period $2\pi/\omega$ (shorter than 2π), rather than with period 2π.
- The particular integral is simply a multiple of $\cos t$, with no factor of t, so there is no term like $t\sin t$.

- Hence x (and likewise $y = \frac{1}{5}(\dot{x} + 2x)$) is a sum of sinusoids of frequencies 1 and ω. Both stay bounded for all t, instead of oscillating with an amplitude that grows linearly in t as when $a = 1$. The motion is periodic only when ω is rational; in general it is a superposition of two oscillations that "beat" against each other.



Checks

- Substituting $x = 5(t-1)\sin t$ and $y = (t-1)\cos t + (2t-1)\sin t$ back into both original equations works identically. A step-by-step numerical solution (fourth-order Runge–Kutta from $x = 0$, $y = -1$) agrees with the formulas to within 10^{-10} on $0 \le t \le 10$; for example $x(10) = 45\sin 10 \approx -24.481$.
- For $a > 1$ the full solution turns out to be $x = \frac{2}{a-1}(\cos t - \cos \omega t) - \frac{5}{\omega}\sin \omega t$. As $a \to 1$ we have $\omega - 1 \approx \frac{5}{2}(a-1)$ and $\cos t - \cos \omega t \approx (\omega - 1)\, t \sin t$, which recovers $5t\sin t - 5\sin t$. The resonant answer is the limit of the non-resonant one.

Common slips

- Using $y(0) = 0$ instead of deducing $y(0) = -1$ from the condition on $\dot{y}(0)$.
- Trying $C\cos t + D\sin t$ as the particular integral when $a = 1$: it is part of the complementary function, so it yields $0 = 10\cos t$.
- Solving separately for y with new arbitrary constants: once x is known, y is fixed by $y = \frac{1}{5}(\dot{x} + 2x)$.

Beyond STEP

A linear system like this is really $\dot{\mathbf{v}} = M\mathbf{v} + \mathbf{f}(t)$ with $\mathbf{v} = (x, y)$ and a constant matrix M; the eigenvalues of M, here $\pm\sqrt{4 - 5a}$, decide whether the free motion grows, decays or oscillates. Resonance happens when the forcing frequency matches an eigenvalue $\pm$i, which is exactly $a = 1$. Boyce and DiPrima, *Elementary Differential Equations*, treats both systems via matrices and forced oscillations with resonance and beats.

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